

hovid-Valsar C Film-coated Tablet 160mg / 12.5mg

VIVALxx-PC

DESCRIPTION

Oblong, dark red film-coated tablet, shallow convex, bevel-edged, break bar on one face and 'HOVID' embossed on the other face.

COMPOSITION

Each tablet contains:
Valsartan 160 mg
Hydrochlorothiazide 12.5 mg

ACTIONS AND PHARMACOLOGY

Angiotensin II antagonist.

The active hormone of the renin-angiotensin-aldosterone system (RAAS) is angiotensin II, which is formed from angiotensin I through ACE. Angiotensin II binds to specific receptors located in the cell membranes of various tissues. It has a wide variety of physiological effects, including in particular both direct and indirect involvement in the regulation of blood pressure. As a potent vasoconstrictor, angiotensin II exerts a direct pressor response. In addition, it promotes sodium retention and stimulation of aldosterone secretion.

Valsartan is an orally active, potent, and specific angiotensin II (Ang II) receptor antagonist. It acts selectively on the AT1 receptor subtype, which is responsible for the known actions of angiotensin II. The increased plasma levels of Ang II following AT1 receptor blockade with valsartan may stimulate the unblocked AT2 receptor, which appears to counterbalance the effect of the AT1 receptor. Valsartan does not exhibit any partial agonist activity at the AT1 receptor and has much (about 20,000 fold) greater affinity for the AT1 receptor than for the AT2 receptor.

Valsartan does not inhibit ACE (also known as kinase II) which converts Ang I to Ang II and degrades bradykinin. Since there is no effect on ACE and no potentiation of bradykinin or substance P, angiotensin II antagonists are unlikely to be associated with cough. Valsartan does not bind to or block other hormone receptors or ion channels known to be important in cardiovascular regulation.

Administration of valsartan to patients with hypertension results in reduction of blood pressure without affecting pulse rate.

In most patients, after administration of a single oral dose, onset of antihypertensive activity occurs within 2 hours, and the peak reduction of blood pressure is achieved within 4-6 hours. The antihypertensive effect persists over 24 hours after dosing. During repeated dosing, the maximum reduction in blood pressure with any dose is generally attained within 2-4 weeks and is sustained during long term therapy. Combined with hydrochlorothiazide, a significant additional reduction in blood pressure is achieved.

Abrupt withdrawal of valsartan has not been associated with rebound hypertension or other adverse clinical events.

The site of action of thiazide diuretics is primarily in the renal distal convoluted tubule. It has been shown that there is a high-affinity receptor in the renal cortex as the primary binding site for the thiazide diuretic action and inhibition of NaCl transport in the distal convoluted tubule. The mode of action of thiazides is through inhibition of the Na⁺/Cl⁻ symporter perhaps by competing for the Cl⁻ site, thereby affecting electrolyte reabsorption mechanisms directly increasing sodium and chloride excretion to an approximately equal extent, and indirectly by this diuretic action reducing plasma volume, with consequent increases in plasma renin activity, aldosterone secretion and urinary potassium loss, and a decrease in serum potassium. The renin-aldosterone link is mediated by angiotensin II, so with co-administration of valsartan the reduction in serum potassium is less pronounced as observed under monotherapy with hydrochlorothiazide.

PHARMACOKINETICS

Valsartan

Absorption of valsartan after oral administration is rapid, although the amount absorbed varies widely. Mean absolute bioavailability for valsartan is 23%. Valsartan shows multiexponential decay kinetics ($t_{1/2}$ alpha < 1 hour and $t_{1/2}$ beta about 9 hours).

The pharmacokinetics of valsartan are linear in the dose range tested. There is no change in the kinetics of valsartan on repeated administration, and little accumulation when dosed once daily. Plasma concentrations were observed to be similar in males and females.

Valsartan is highly bound to serum protein (94-97%), mainly serum albumin. Steady-state volume of distribution is low (about 17 L).

Plasma clearance is relatively slow (about 2 L/h) when compared with hepatic blood flow (about 30 L/h). Of the absorbed dose of valsartan, 70% is excreted in the faeces and 30% in the urine, mainly as unchanged compound.

When valsartan is given with food, the area under the plasma concentration curve (AUC) of valsartan is reduced by 48%, although from about 8 hours post dosing, plasma valsartan concentrations are similar for the fed and fasted group. However, this reduction in AUC is not accompanied by a clinically significant reduction in the therapeutic effect, and valsartan can therefore be given either with or without food.

The average time to peak concentration and elimination half-life of valsartan in heart failure patients are similar to that observed in healthy volunteers. AUC and C_{max} values of valsartan increase linearly and are almost proportional with increasing dose over the clinical dosing range (40 to 160 mg twice a day). The average accumulation factor is about 1.7. The apparent clearance of valsartan following oral administration is approximately 4.5 L/h. Age does not affect the apparent clearance in heart failure patients.

Hydrochlorothiazide

Absorption: The absorption of hydrochlorothiazide, after an oral dose, is rapid (t_{max} about 2 h). The increase in mean AUC is linear and dose proportional in the therapeutic range.

The effect of food on hydrochlorothiazide absorption, if any, has little clinical significance. Absolute bioavailability of hydrochlorothiazide is 70% after oral administration.

Distribution: The apparent volume of distribution is 4-8 l/kg.

Circulating hydrochlorothiazide is bound to serum proteins (40-70%), mainly serum albumin. Hydrochlorothiazide also accumulates in erythrocytes at approximately 3 times the level in plasma.

Elimination: Hydrochlorothiazide is eliminated predominantly as unchanged drug. Hydrochlorothiazide is eliminated from plasma with a half-life averaging 6 to 15 hours in the terminal elimination phase. There is no change in the kinetics of hydrochlorothiazide on repeated dosing, and accumulation is minimal when dosed once daily. There is more than 95% of the absorbed dose being excreted as unchanged compound in the urine. The renal clearance is composed of passive filtration and active secretion into the renal tubule.

Valsartan / Hydrochlorothiazide

The systemic availability of hydrochlorothiazide is reduced by about 30% when coadministered with valsartan. The kinetics of valsartan are not markedly affected by the coadministration of hydrochlorothiazide. This observed interaction has no impact on the combined use of valsartan and hydrochlorothiazide, since controlled clinical trials have shown a clear antihypertensive effect, greater than that obtained with either active substance given alone, or placebo.

Elderly: A somewhat higher systemic exposure to valsartan was observed in some elderly subjects than in young subjects; however, this has not been shown to have any clinical significance.

Limited data suggest that the systemic clearance of hydrochlorothiazide is reduced in both healthy and hypertensive elderly subjects compared to young healthy volunteers.

Renal impairment: At the recommended dose of hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg no dose adjustment is required for patients with a Glomerular Filtration Rate (GFR) of 30-70 ml/min. In patients with severe renal impairment (GFR < 30 ml/min) and patients undergoing dialysis no data are available for hovid-Valsar C Film-coated Tablet 160 mg / 12.5 mg. Valsartan is highly bound to plasma protein and is not to be removed by dialysis, whereas clearance of hydrochlorothiazide will be achieved by dialysis.

In the presence of renal impairment, mean peak plasma levels and AUC values of hydrochlorothiazide are increased and the urinary excretion rate is reduced. In patients with mild to moderate renal impairment, a 3-fold increase in hydrochlorothiazide AUC has been observed. In patients with severe renal impairment an 8-fold increase in AUC has been observed. Hydrochlorothiazide is contraindicated in patients with severe renal impairment.

Hepatic impairment: In a pharmacokinetics trial in patients with mild (n=6) to moderate (n=5) hepatic dysfunction, exposure to valsartan was increased approximately 2-fold compared with healthy volunteers.

There is no data available on the use of valsartan in patients with severe hepatic dysfunction. Hepatic disease does not significantly affect the pharmacokinetics of hydrochlorothiazide.

INDICATION

Treatment of hypertension.

hovid-Valsar C Film-coated Tablet 160mg/12.5mg fixed-dose combination is indicated in patients whose blood pressure is not adequately controlled on valsartan or hydrochlorothiazide monotherapy.

CONTRAINDICATIONS

- Hypersensitivity to valsartan, hydrochlorothiazide, other sulfonamide-derived medicinal products or to any of the excipients.
- Second and third trimester of pregnancy
- Severe hepatic impairment, biliary cirrhosis and cholestasis.
- Severe renal impairment (creatinine clearance < 30 ml/min), anuria.
- Refractory hypokalaemia, hyponatraemia, hypercalcaemia, and symptomatic hyperuricaemia.
- Concomitant use of angiotensin receptor antagonists (ARBs) - including valsartan - or of angiotensin-converting-enzyme inhibitors (ACEIs) with aliskiren in patients with diabetes mellitus or renal impairment (GFR < 60 mL/min/1.73m²)

PREGNANCY AND LACTATION

Pregnancy

Valsartan: The use of Angiotensin II Receptor Antagonists (AIIRAs) is not recommended during first trimester of pregnancy. The use of AIIRAs is contra-indicated during the second and third trimester of pregnancy.

Epidemiological evidence regarding the risk of teratogenicity following exposure to ACE inhibitors during the first trimester of pregnancy has not been conclusive; however a small increase in risk cannot be excluded. Whilst there is no controlled epidemiological data on the risk with Angiotensin II Receptor Inhibitors (AIIRAs), similar risks may exist for this class of drugs. Unless continued AIIRAs therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with AIIRAs should be stopped immediately and, if appropriate, alternative therapy should be started.

AIIRAs therapy exposure during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalaemia).

Should exposure to AIIRAs have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.

Infants whose mothers have taken AIIRAs should be closely observed for hypotension.

Hydrochlorothiazide: There is limited experience with hydrochlorothiazide during pregnancy, especially during the first trimester. Hydrochlorothiazide crosses the placenta. Based on the pharmacological mechanism of action of hydrochlorothiazide its use during the second and third trimester may compromise foeto-placental perfusion and may cause foetal and neonatal effects like icterus, disturbance of electrolyte balance and thrombocytopenia.

Lactation

No information is available regarding the use of valsartan during breastfeeding. Hydrochlorothiazide is excreted in human milk. Therefore the use of hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg during breast feeding is not recommended. Alternative treatments with better established safety profiles during breast-feeding are preferable, especially while nursing a newborn or preterm infant.

PRECAUTIONS

Serum electrolyte changes: Concomitant use with potassium supplements, potassium-sparing diuretics, salt substitutes containing potassium, or other drugs that may increase potassium levels (heparin, etc.) is not recommended.

Hypokalaemia has been reported under treatment with thiazide diuretics. Frequent monitoring of serum potassium is recommended.

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Treatment with thiazide diuretics, including hydrochlorothiazide has been associated with hyponatraemia and hypochloremic alkalosis. Thiazides, including hydrochlorothiazide increase the urinary excretion of magnesium, which may result in hypomagnesaemia. Calcium excretion is decreased by thiazide diuretics. This may result in hypercalcaemia.

- **Sodium, and/or volume-depleted patients:** In severely sodium-depleted and/or volume-depleted patients, such as those receiving high doses of diuretics, symptomatic hypotension may occur in rare cases after initiation of therapy with hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg. Sodium and/or volume depletion should be corrected before starting treatment with hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg.
- **Renal artery stenosis:** hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg should not be used to treat hypertension in patients with unilateral or bilateral renal artery stenosis or stenosis of the artery to a solitary kidney, since blood urea and serum creatinine may increase in such patients.
- **Renal impairment:** No dosage adjustment is required for patients with renal impairment (creatinine clearance >30 mL/min).
- **Hepatic impairment:** In patients with mild to moderate hepatic impairment without cholestasis, hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg should be used with caution. Thiazides should be used with caution in patients with impaired hepatic function or progressive liver disease, since minor alterations of fluid and electrolyte balance may precipitate hepatic coma.
- **Systemic lupus erythematosus:** Thiazide diuretics, including hydrochlorothiazide, have been reported to exacerbate or activate systemic lupus erythematosus.
- **Other metabolic disturbances:** Thiazide diuretics, including hydrochlorothiazide, may alter glucose tolerance and raise serum levels of cholesterol, triglycerides and uric acid.

INTERACTIONS WITH OTHER MEDICAMENTS

Reversible increases in serum lithium concentrations and toxicity have been reported during concurrent use of ACE inhibitors and thiazide, including hydrochlorothiazide. Due to the lack of experience with concomitant use of valsartan and lithium, this combination is not recommended. If the combination proves necessary, careful monitoring of serum lithium levels is recommended.

hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg may increase the effects of other agents with antihypertensive properties (e.g. guanethidine, methyldopa, vasodilators, ACEI, ARBs, beta-blockers, calcium channel blockers and DRIs).

Non-steroidal anti-inflammatory medicines (NSAIDs), including selective COX-2 inhibitors, acetylsalicylic acid (>3 g/day), and non-selective NSAIDs: NSAIDs can attenuate the antihypertensive effect of both angiotensin II antagonists and hydrochlorothiazide when administered simultaneously. Furthermore, concomitant use of hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg and NSAIDs may lead to worsening of renal function and an increase in serum potassium. Therefore, monitoring of renal function at the beginning of the treatment is recommended, as well as adequate hydration of the patient.

Dual blockade of the Renin-Angiotensin-System (RAS) with ARBs, ACEIs, or aliskiren: Caution is required while co-administering ARBs, including valsartan, with other agents blocking the RAAS such as ACEIs or aliskiren. Concomitant use of angiotensin receptor antagonists (ARBs) – including valsartan – or of angiotensin-converting-enzyme inhibitors (ACEIs) with aliskiren in patients with diabetes mellitus or renal impairment (GFR < 60 mL/min/1.73m²) is contraindicated.

Potassium-sparing diuretics, potassium supplements, salt substitutes containing potassium and other substances that may increase potassium levels: If a medicinal product that affects potassium levels is considered necessary in combination with valsartan, monitoring of potassium plasma levels is advised.

Medicinal products affecting serum potassium level: The hypokalaemic effect of hydrochlorothiazide may be increased by concomitant administration of kaliuretic diuretics, corticosteroids, laxatives, ACTH, amphotericin, carbenoxolone, penicillin G, salicylic acid and derivatives.

If these medicinal products are to be prescribed with the hydrochlorothiazide-valsartan combination, monitoring of potassium plasma levels is advised.

Medicinal products affecting serum sodium level: The hyponatraemic effect of diuretics may be intensified by concomitant administration of drugs such as antidepressants, antipsychotics, antiepileptics, etc. Caution is advised in long-term administration of these drugs.

Digitalis glycosides: Thiazide-induced hypokalaemia or hypomagnesaemia may occur as undesirable effects favouring the onset of digitalis-induced cardiac arrhythmias.

Calcium salts and vitamin D: Administration of thiazide diuretics, including hydrochlorothiazide, with vitamin D or with calcium salts may potentiate the rise in serum calcium. Concomitant use of thiazide type diuretics with calcium salts may cause hypercalcaemia in patients pre-disposed for hypercalcaemia (e.g. hyperparathyroidism, malignancy or vitamin-D-mediated conditions) by increasing tubular calcium reabsorption.

Antidiabetic agents (oral agents and insulin): Thiazides may alter glucose tolerance. Dose adjustment of the antidiabetic medicinal product may be necessary. Metformin should be used with caution because of the risk of lactic acidosis induced by possible functional renal failure linked to hydrochlorothiazide.

Beta blockers and diazoxide: Concomitant use of thiazide diuretics, including hydrochlorothiazide, with beta blockers may increase the risk of hyperglycaemia. Thiazide diuretics, including hydrochlorothiazide, may enhance the hyperglycaemic effect of diazoxide.

Medicinal products used in the treatment of gout (probenecid, sulfapyrazone and allopurinol): Dose adjustment of uricosuric medications may be necessary as hydrochlorothiazide may raise the level of serum uric acid. Increase of dosage of probenecid or sulfapyrazone may be necessary. Co-administration of thiazide diuretics, including hydrochlorothiazide, may increase the incidence of hypersensitivity reactions to allopurinol.

Anticholinergic agents and other medicinal products affecting gastric motility: The bioavailability of thiazide-type diuretics may be increased by anticholinergic agents (e.g. atropine, biperiden), apparently due to a decrease in gastrointestinal motility and the stomach emptying rate. Conversely, it is anticipated that prokinetic drugs such as cisapride may decrease the bioavailability of thiazide-type diuretics.

Amantadine: Thiazides, including hydrochlorothiazide, may increase the risk of adverse effects caused by amantadine.

Cytotoxic agents: Thiazides, including hydrochlorothiazide, may reduce renal excretion of cytotoxic agents (e.g. cyclophosphamide, methotrexate) and potentiate their myelosuppressive effects.

Non-depolarising skeletal muscle relaxants (e.g. tubocurarine): Thiazides, including hydrochlorothiazide, potentiate the action of skeletal muscle relaxants such as curare derivatives.

Ciclosporin: Concomitant treatment with ciclosporin may increase the risk of hyperuricaemia and gout-type complications.

Methyldopa: There have been isolated reports of haemolytic anaemia in patients receiving concomitant treatment with methyldopa and hydrochlorothiazide.

MAIN SIDE/ADVERSE EFFECTS

Adverse reactions reported in clinical trials and laboratory findings occurring more frequently with valsartan plus hydrochlorothiazide versus placebo and individual postmarketing reports are presented below according to system organ class. Adverse reactions known to occur with each component given individually but which have not been seen in clinical trials may occur during treatment with valsartan/hydrochlorothiazide.

Frequency of adverse reactions with valsartan / hydrochlorothiazide

- **Uncommon:** Dehydration, Paraesthesia, Vision blurred, Tinnitus, Hypotension, Cough, Myalgia, Fatigue
- **Very rare:** Dizziness, Diarrhoea, Arthralgia

Frequency of adverse reactions with valsartan

- **Uncommon:** Vertigo, Abdominal pain

Frequency of adverse reactions with hydrochlorothiazide

Hydrochlorothiazide has been extensively prescribed for many years, frequently in higher doses than those administered with hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg.

The following adverse reactions have been reported in patients treated with monotherapy of thiazide diuretics, including hydrochlorothiazide:

- **Very common:** Hypokalaemia, blood lipids increased (mainly at higher doses)
- **Common:** Hyponatraemia, hypomagnesaemia, hyperuricaemia, Postural hypotension, Loss of appetite, mild nausea and vomiting, Urticaria and other forms of rash, Impotence
- **Rare:** Thrombocytopenia sometimes with purpura, Hypercalcaemia, hyperglycaemia, glycosuria and worsening of diabetic metabolic state, Depression, sleep disturbances, Headache, dizziness, paraesthesia, Visual impairment, Cardiac arrhythmias, Constipation, gastrointestinal discomfort, diarrhoea, Intrahepatic cholestasis or jaundice, Photosensitisation
- **Very rare:** Agranulocytosis, leucopenia, haemolytic anaemia, bone marrow failure, Hypersensitivity reactions, Hypochloremic alkalosis, Respiratory distress including pneumonitis and pulmonary oedema, Pancreatitis, Necrotising vasculitis and toxic epidermal necrolysis, cutaneous lupus erythematosus-like reactions, reactivation of cutaneous lupus erythematosus

OVERDOSE AND TREATMENT

Overdose with valsartan may result in marked hypotension, which could lead to depressed level of consciousness, circulatory collapse and/or shock. In addition, the following signs and symptoms may occur due to an overdose of the hydrochlorothiazide component: nausea, somnolence, hypovolaemia, and electrolyte disturbances associated with cardiac arrhythmias and muscle spasms.

If hypotension occurs, the patient should be placed in the supine position and salt and volume supplementation should be given rapidly. Valsartan cannot be eliminated by means of haemodialysis because of its strong plasma binding behaviour whereas clearance of hydrochlorothiazide will be achieved by dialysis.

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DOSAGE AND ADMINISTRATION

For oral use.
hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg can be taken with or without food and should be administered with water.

The recommended dose of hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg is one film-coated tablet once daily. Dose titration with the individual components is recommended. In each case, up-titration of individual components to the next dose should be followed in order to reduce the risk of hypotension and other adverse events.

The antihypertensive effect is substantially present within 2 weeks. In most patients, maximal effects are observed within 4 weeks. However, in some patients, 4-8 weeks treatment may be required. This should be taken into account during dose-titration.

No dose adjustment is required for patients with mild to moderate renal impairment (Glomerular Filtration Rate (GFR) > 30 mL/min). Due to the hydrochlorothiazide component, hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg is contraindicated in patients with severe renal impairment (GFR < 30 mL/min) and anuria. Concomitant use of valsartan with aliskiren is contraindicated in patients with renal impairment (GFR < 60 mL/min/1.73 m²).

In patients with mild to moderate hepatic impairment without cholestasis the dose of valsartan should not exceed 80 mg. No adjustment of the hydrochlorothiazide dose is required for patients with mild to moderate hepatic impairment.

No dose adjustment is required in elderly patients.

hovid-Valsar C Film-coated Tablet 160 mg/12.5 mg is not recommended for use in children below the age of 18 years due to a lack of data on safety and efficacy.

Storage: Store below 30°C.
Presentation / Packing: Blister of 3 x 10's

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